A-NUCA-DREM for Energy-Aware Autonomous EV Platooning under Mixed Traffic Conditions

Abstract:

In addition to modelling of vehicle activity and congestion, the fast evolving autonomous and electric vehicle landscape necessitates the modelling of heterogeneous driving behaviour, vehicle-to-vehicle interaction, platoon formation, traffic-density variation and energy-related vehicle state. Although conventional cellular automata (CA) models are a computationally efficient representation of traffic flow, the transition rules are uniform and mainly reactive, restricting their representation of the adaptive vehicle behaviour. This project aims to create an Adaptive Non-Uniform Cellular Automata with Dynamic Rule Evolution Mechanism (A-NUCA-DREM) framework to model energy-efficient autonomous electric-vehicle (EV) traffic and platooning in mixed traffic. The development is executed step-by-step in five computational notebooks: from a classical Nagel–Schreckenberg (NaSch) baseline to more realistic, trajectory-informed adaptive decision making.

The first step is to set up the classical NaSch cellular automata model as a baseline traffic simulator. Discrete cells are used to represent a one-dimensional highway lattice where the main attributes of a vehicle state are its position and speed. The four standard NaSch operations (acceleration, deceleration based on a given gap, stochastic randomization and movement) are used to model basic traffic phenomena. Implementation produces vehicle trajectories, space–time diagrams, speed statistics, density statistics, fundamental diagrams, traffic animation and congestion related behaviours. This baseline is a proof of the capability of classical CA to simulate free-flow traffic, congestion formation, stop and go and traffic shocks. Concurrently, it points to some key constraints: all vehicles use the same transition mechanism, no attention is paid to vehicle heterogeneity, energy state is not involved in decision making, platooning is not dynamically supported, future traffic conditions are not predicted, and the traffic rules are fixed.

The second stage presents the proposed Adaptive Non-Uniform Cellular Automata (A-NUCA) and Dynamic Rule Evolution Mechanism (DREM). In addition to position and velocity, the adaptive state also embodies power/energy related information, State of Charge (SOC), local traffic density, predicted density, remaining energy, platoon state, and neighbourhood relationships. The framework contains eight adaptable rules that can be interpreted as the following: R1—free acceleration, R2—platoon joining, R3—platoon maintaining, R4—increasing headway, R5—platoon splitting, R6—lane change, R7—eco-driving during lane change, and R8—cooperative braking. These rules allow for vehicle response to local traffic and vehicle state to vary. The resulting adaptive cycle is as follows: $S_i(t) \rightarrow N_i \rightarrow \rho_i \rightarrow \hat{\rho}_i \rightarrow R_i \rightarrow F_i \rightarrow S_i(t+1)$, with the neighbourhood information and vehicle traffic-density states influencing the choice of rules, and the resulting rules influencing the vehicle-state transition.

The third stage is an extension of the framework that adds an advanced EV battery and simulation-logging component. The aim of the goal is to relate traffic behaviour to energy-related vehicle states, not to deal with traffic flow independently. The model calculates the

power demand of the vehicles, the current of the battery, the temperature of the battery, the cumulative energy used, and the degradation quantities of the battery, as the time passes, recording the state of the simulations. Additionally, Platoon and traffic analytics are integrated, to enable measures at vehicle-level and system-level. This establishes a unified computational platform, enabling the relation between changes in acceleration and speed, platoon behaviour and traffic conditions to EV energy consumption and SOC.

The fourth stage adds real world traffic information via NGSIM trajectory data from US-101 and I-80. Data-processing steps are loading, cleaning, standardizing, converting units, reconstructing trajectories, analysing vehicle dynamics, analysing lane changes, computing travel distance, computing travel time and estimating local density. The processed trajectories are then converted to A-NUCA adaptive states and from these states, the rule decision information is prepared. It is important to note that throughout the framework, NGSIM provides traffic information (position, velocity, acceleration, lane, headway, vehicle relationships) observed in real traffic, while EV states (such as SOC, battery energy, and battery temperature) are estimated or modelled. So, any energy-related outputs are not directly measured EV battery data from NGSIM, but rather the behaviour of the proposed EV-state model when paired with actual traffic patterns.

The fifth stage is the final piece of the adaptive prediction and decision pipeline. The level of traffic density is obtained from the NGSIM trajectories and transformed into a time series at the frame level. Lagged density features are used to train a compact multilayer perceptron (MLP) regression model for predicting future traffic density. The evaluation results of the prediction data used in the implemented prediction model show an MAE of 0.001969, RMSE of 0.007837, and R^2 of 0.832361, which reflects a real predicted-density signal for the adaptive framework. The density is then assumed to be the predicted density and fed into the A-NUCA state, and then delivered to DREM via the rule-selection relationship $R_i = \Phi(SOC_i, \rho_i, P_i, N_i, \hat{\rho}_i)$. All vehicles are compared to the eight-rule library and the chosen rule is applied to create the next adaptive state. Update the transition: acceleration, velocity, position, lane (if applicable), energy consumption, remaining energy, and SOC.

Final evaluation was conducted on an integrated data set of 1298 vehicles and 7180 frames. The evaluation at the A-NUCA reported the average speed as 7.5092, the mean absolute acceleration as 0.5359, the average SOC as 100.0%, the total modelled energy consumption as 15,965.7792, and the average density as 31.8840, the average predicted density as 27.0241. The final evaluation MAE and the RMSE for the density comparison were 4.8599 and 12.6337, respectively. The analysis of the distribution rules revealed that the most common rule selected was R3 (platoon maintaining) with 57.63% of the evaluated states, followed by R8 (cooperative braking) with 18.57%, R1 (acceleration) 18.38%, and R4 (headway increase) 5.41%. Under the implemented definition of platoon, the evaluated set of data revealed 100% engagement of recorded states for platoon states. Mean energy consumption and post transition speed statistics were also calculated on a rule-wise basis, enabling the individual adaptive behaviours to be analysed.

Overall, it is a five-stage development that forms a comprehensive traffic model trajectory from classical reactive traffic to predictive, state-aware, rule-adaptive traffic. The main contribution lies in the coupling of the cellular automata traffic dynamics with heterogeneous vehicle states, dynamic rule evolution, EV energy modelling, platoon behaviour, real-world vehicle trajectories, prediction of traffic density, and adaptive states transition in one computational architecture. The framework will serve as a basis for further research with more sophisticated models of EV batteries, experimentally measured energy values, more complex prediction architectures, multi-lane cooperative decision making, and systematic comparison with traditional CA and other autonomous-driving control approaches.

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